Import Data Using Transform Maps (Spreadsheet)

Phase 2 - Requirement Analysis

1. Requirement Overview

The purpose of this project is to import data from a spreadsheet into a ServiceNow target table using **Import Sets and Transform Maps**.

The system should allow users to upload spreadsheet data, temporarily store the imported data, map spreadsheet fields to ServiceNow fields, transform the data when required, validate the information, and finally insert or update records in the target table.

The requirement analysis focuses on identifying the **functional requirements, non-functional requirements, data requirements, validation requirements, and error-handling requirements** of the import process.

2. Business Requirement

The organization requires a simple and reliable mechanism to transfer bulk data from spreadsheets into ServiceNow.

The solution should:

- Reduce manual data-entry work.
- Support bulk data import.
- Maintain data accuracy and consistency.
- Map spreadsheet columns to appropriate ServiceNow fields.
- Transform source data according to business requirements.
- Prevent unnecessary duplicate records.
- Identify invalid or incomplete records.
- Provide visibility into successful and failed imports.

3. Functional Requirements

3.1 Spreadsheet Upload

The system should allow users to upload a spreadsheet containing the required data.

The spreadsheet may contain fields such as:

- Employee ID
- Employee Name
- Email
- Department
- Phone Number
- Joining Date

The uploaded file should contain appropriate column headers and data in the expected format.

3.2 Import Set Requirement

The uploaded spreadsheet data should first be loaded into an **Import Set Table**.

The Import Set Table acts as a temporary staging area between the spreadsheet and the final ServiceNow target table.

Flow:

Spreadsheet → Import Set Table → Transform Map → Target Table

3.3 Transform Map Requirement

A Transform Map should be created to define the relationship between the source fields and target fields.

For example:

Source Field	Target Field
Employee ID	Employee ID
Employee Name	Name
Email	Email
Department	Department
Joining Date	Date of Joining

ss to handle both new and existing data.

6. Data Validation Requirements

Before the data is inserted into the target tThe Transform Map should ensure that data is transferred to the correct target fields.

3.4 Field Mapping Requirement

Each required spreadsheet column should be mapped to its corresponding ServiceNow field.

The mapping should consider:

- Field name
- Field type
- Mandatory status
- Data format
- Reference fields
- Transformation requirements

Unnecessary spreadsheet columns should not be transferred to the target table unless they are required by the business process.

4. Data Transformation Requirements

The system should support transformation of source data when the spreadsheet format differs from the target table format.

Examples include:

- Converting date formats.
- Removing unnecessary spaces.
- Standardizing text values.
- Converting numeric values.
- Mapping source values to reference records.
- Handling blank values.
- Applying business-specific transformation rules.

For example, if the spreadsheet contains:

```
IT Department
```

but the target field expects a specific department reference, the Transform Map should identify and map the value appropriately.

5. Duplicate Record Requirement

The system should prevent unnecessary duplicate records.

A unique field such as **Employee ID** can be configured as a **Coalesce field**.

The requirement is:

Existing Employee ID → Update existing record

New Employee ID → Create new record

This allows the import proceable, the system should validate important fields.

Validation requirements may include:

- Employee ID should not be empty.
- Employee Name should contain valid data.
- Email should follow a valid format.
- Department should contain an acceptable value.

- Date fields should follow the expected format.
- Mandatory fields should be populated.
- Duplicate values should be identified where applicable.

Invalid records should not be allowed to create incorrect target data.

7. Error Handling Requirements

The system should provide appropriate error handling during the import and transformation process.

The system should be able to identify:

- Missing mandatory fields.
- Invalid field values.
- Incorrect data formats.
- Invalid reference values.
- Duplicate records.
- Transformation failures.
- Mapping errors.

Failed records should be identifiable so that users can correct the spreadsheet data and perform the import again.

8. Target Table Requirements

The target ServiceNow table should contain all necessary fields required by the business process.

For example:

Field	Requirement
Employee ID	Mandatory / Unique
Name	Mandatory
Email	Mandatory
Department	Required

Field	Requirement
Phone	Optional
Date of Joining	Required

The target table should be configured before the Transform Map is executed.

9. User Requirements

The user should be able to:

1. Prepare the spreadsheet according to the required format.
2. Upload the spreadsheet.
3. Start the import process.
4. Execute the Transform Map.
5. Review imported records.
6. Identify failed records.
7. Correct invalid data.
8. Re-import the corrected data.
9. Verify the final target records.

10. Non-Functional Requirements

Performance

- The system should support bulk data import efficiently.
- The import process should be completed within a reasonable time depending on the data volume.

Accuracy

- Source data should be mapped to the correct target fields.
- Transformation rules should produce consistent results.

Reliability

- The import process should handle valid and invalid records appropriately.
- Failed transformations should be identifiable.

Security

- Only authorized users should be allowed to perform data imports.
- Imported data should follow ServiceNow access and security rules.

Maintainability

- Transform Maps should be easy to understand and maintain.
- Field mappings and transformation logic should be documented.
- Changes to the spreadsheet structure should be manageable.

11. Assumptions

The following assumptions are considered during requirement analysis:

- The spreadsheet contains valid column headers.
- The user has permission to import data into ServiceNow.
- The target table already exists.
- Required target fields are properly configured.
- The data volume is within the supported import capacity.
- Unique identifiers are available for duplicate checking.
- Users will follow the defined spreadsheet template.

12. Constraints

Some possible constraints are:

- Incorrect spreadsheet formats may cause import failures.
- Missing mandatory fields can prevent successful transformation.
- Large data volumes may increase processing time.
- Reference fields may require additional transformation logic.
- Changes in the target table structure may require changes to the Transform Map.

13. Requirement Traceability

Requirement	Solution Component
Upload spreadsheet	Import Set
Temporarily store source data	Import Set Table
Map source to target fields	Transform Map
Transform data	Field Mapping / Transform Script
Avoid duplicates	Coalesce
Validate data	Transform/Validation logic
Handle errors	Import/Transform error handling

Requirement	Solution Component
Store final data	Target ServiceNow Table
Verify imported records	Target Table / Import Results

14. Overall Requirement Flow

The complete requirement can be represented as:

User prepares Spreadsheet
↓
Upload Spreadsheet
↓
Create/Load Import Set
↓
Store Data in Import Set Table
↓
Apply Transform Map
↓
Map Source Fields to Target Fields
↓
Transform & Validate Data
↓
Check Existing Records using Coalesce
↓
Insert New / Update Existing Records
↓
Handle Errors and Failed Records
↓
Verify Target Records

15. Final Requirement Statement

The system shall provide a mechanism to import spreadsheet data into a ServiceNow target table using Import Sets and Transform Maps. The system shall support field mapping, data

transformation, validation, duplicate handling, error management, and insertion or updating of target records. The solution should reduce manual data entry, improve data accuracy, and provide a reliable process for bulk data migration.